



Célian DI GIOVANNI

Profile

Final-year Engineering Student (M.Sc.) specializing in **Artificial Intelligence applied to physiological signals and clinical data**. Experience with real-world ECG data, biometric classification, medical image segmentation, application validation in a hospital environment, and clinical data structuring.

Seeking a 6-month final-year internship starting April 1st, 2026.

Information

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- B
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- celian06.github.io

Skills

- Python
- AI & Data Science: Classification, Deep Learning (CNN), Feature Engineering, Embedded AI
- Signal Processing: ECG, FFT, Bioimpedance
- Computer Vision: Segmentation, Otsu's Thresholding, Sobel, Mathematical Morphology
- Biostatistics
- Neurosciences
- Java / C

Languages

- English - B2 Cambridge
- Spanish

Formation

Digital and Electronic Engineering (M.Sc.) – e-Health Option	2023 – Present
ISEN Méditerranée, Toulon	
Bionics Engineering – Erasmus Semester	sept. 2024
Università di Pisa	
CPGE PCSI / PC	2021 – 2023
Lycée International de Valbonne	
French Scientific Baccalaureate (Mathematics & Physics-Chemistry) - With Honors	2021
Institution du Mont Saint-Jean, Antibes	

Projects

1st Prize at I-Novgames 2026 - Reinventing Patient Monitoring via Embedded Technology	2026
Carelink Project - Anticipation and Detection of Epileptic Seizures	
- Analyzed ECG signals from the PhysioNet (CHB-MIT) database. - Computed heart rate variability indicators: RMSSD, SDNN, LF/HF. - Trained TensorFlow Lite AI models using Edge Impulse. - Converted and compressed models for execution on STM32 targets. - Processed data from 10 patients and ~1000 30-second windows.	
Exploratory Results: Detected ECG alterations consistent with documented seizure signatures. Global Accuracy: 92% . Functional Proof of Concept for a real-time anticipation and alert system.	
Skin Maven Bandage	2025
Biometric Classification via AI (Master 1 Project)	
- Built a database using skin impedance measurements collected from 83 volunteers. - Applied data augmentation using SMOTE and randomization. - Trained and compared multiple models: Logistic Regression, Random Forest, Gaussian, KNN, SVC.	

Results: Random Forest: 90% accuracy for gender, 60% for age, 20% for skin tone.

Interests

- Culture: Travel, History.
- Music & Arts: Acoustic and Electric Guitar, Cinema, Rock Music.
- Sports: Paddleboarding, Judo, Badminton, Running.
- Volunteering: Order of Malta Outreach (Social Patrols).

Certificates

- First Aid Certification (Level 1)

Region of Interest (ROI) Segmentation on X-ray Images

2025

- Full implementation in **Python and Octave**.
- Methods applied: Otsu's thresholding, RGB/Grayscale conversion, erosion, dilation.
- Data processed: Brain and Chest X-rays (JPG format).

Result: Functional and validated segmentation algorithm.

Professional Experience

Engineering Intern - QA & Clinical Data

June 2025 - Sept 2025

Nice University Hospital (CHU) - Pharmacy Department

- Performed functional validation of a medical monitoring mobile application (MUSE project).
- Defined **use cases and test scenarios**, and reported bugs/feedback to development teams.
- Acted as a liaison between medical staff and developers.
- Extracted and structured clinical data from unstructured **reports (DOCX/XLSX)**.
- Developed Python automation scripts (using openpyxl, python-docx).
- Centralized and analyzed data using **Excel / Power Query / Pivot Tables**.